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A quick brown fox jumps over the lazy dog

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$$e^{ix} = \cos x + i \sin x$$

$$\left(\sum_{n=0}^{\infty} \frac{1}{n!}\right)^{\sqrt{-64}\left(\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}\right)} = 1$$

fixdif+derivative

$$\int_0^{\pi} \sin x \,\mathrm{d}x = 2$$

$$\frac{\mathrm{d}x^n}{\mathrm{d}x} = nx^{n-1}$$

$$\frac{\mathrm{d}^n x^n}{\mathrm{d}x^n} = n!$$

$$\frac{\mathrm{d}}{\mathrm{d}x} x^n = nx^{n-1}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} x^n = n!$$

$$\frac{\partial x}{\partial f}(x, y)$$

$$\mathrm{d}z = \mathrm{d}f(x, y) = \frac{\partial z}{\partial x}\,\mathrm{d}x + \frac{\partial z}{\partial y}\,\mathrm{d}y = f_x\,\mathrm{d}x + f_y\,\mathrm{d}y$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\int \sin x \,\mathrm{d}x = -\cos x + C$$

$$\mathrm{d}z = \frac{\partial z}{\partial x}\,\mathrm{d}x + \frac{\partial z}{\partial y}\,\mathrm{d}y$$

$$\iint_D f(x, y)\,\mathrm{d}x\,\mathrm{d}y = \iint_{\tilde{D}} f(r, \theta) \cdot r\,\mathrm{d}r\,\mathrm{d}\theta$$